

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 1.1: Pressure — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 1.1: Pressure
Driving Question	DRIVING QUESTION: Why does a sharp knife cut better than a blunt knife when using the same force?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon: Why Does the Sharp Knife Cut but the Blunt Knife Only Squash?	Mama Wanjiru pushes with the same force of 40 N on a sharp knife (edge 0.1 mm thick) and a blunt knife (edge 1 mm thick). The sharp knife slices easily through the tomato and nyama choma; the blunt knife barely cuts and only squashes the food. The person, the force, the food and the size and shape of the knives are the same; the thickness of the edge and the result are different.	The same force can give very different results. Something other than the force must be different between the two knives. Our first ideas point to the edge and the small place where the knife touches the food. We have not yet explained it with evidence, and we have many questions on our Driving Question Board.	Our first model (Version 1, to be tested): the knives get the same push, but the sharp knife has a thinner edge and touches the food in a smaller place, so it cuts, while the blunt knife touches in a bigger place and only squashes the food. We will test and improve this model in the next lessons using evidence.
Lesson 2 What is Pressure? Force and Area on Flour	The same 2 kg brick (weight 20 N) made a deeper dent in the flour when it rested on its small face (10 cm x 5 cm) than on its large face (20 cm x 10 cm).	Pressure depends on both force and area. Pressure is force acting at right angles per unit area: $P = F/A$, measured in pascals (Pa), where 1 Pa = 1 N/m ² . The large, medium and small faces gave 1,000 Pa, 2,000 Pa and 4,000 Pa.	The weight (20 N) was the same on every face, but on the smallest area (0.005 m ²) the force was concentrated, giving the greatest pressure (4,000 Pa) and the deepest dent. This suggests that Mama Wanjiru's sharp knife, with the same force on a much smaller edge area, can give a greater pressure. We will calculate this in Lesson 3.
Lesson 3 How Does Area Affect Pressure? Calculating Pressure for Knives, Camel Feet and Shoes	Same force of 40 N from Mama Wanjiru on both knives, but the contact area differs: sharp knife 5×10^{-6} m ² , blunt knife 5×10^{-5} m ² . Calculated pressures: sharp 8 MPa, blunt 0.8 MPa. Camel (4,000 N, 0.2 m ²) gives 20 kPa, the person (600 N, 0.04	Using $P = F/A$, for a constant force the pressure is larger when the area is smaller (inverse relationship). The sharp edge has 10 times less area and gives 10 times more pressure. A heavier object can give a similar pressure if its weight is spread over	The sharp knife cuts because, with the same force of 40 N, its thin edge has a much smaller area, so the pressure (8 MPa) is greater than the 2 MPa at which the tomato or meat gives way. The blunt knife, with the same force and a larger area, gives

	m ²) gives 15 kPa, and the camel on narrow hooves (0.04 m ²) would give 100 kPa.	a larger area.	only 0.8 MPa, which is less than 2 MPa, so the food is just squashed. The difference is the area, not the force.
Lesson 4 Pressure in Liquids and Depth: The Can with Holes and the Dam Wall	The water from the lowest hole (15 cm) squirted farthest, and the water from the highest hole (5 cm) squirted the least. Pressure values: 500 Pa at 5 cm, 1,000 Pa at 10 cm, 1,500 Pa at 15 cm. A dam with water 30 m deep has 300 kPa at the base compared with 100 kPa at 10 m depth.	Liquids exert pressure on the walls of their container and in all the places they touch, and this pressure increases with depth. More water above a point means more weight pressing there.	Dam walls are built thicker at the bottom because the pressure there (300 kPa at 30 m) is much larger than near the top, so the wall must withstand more. This extends the knife idea: pressure, not just force, decides what a surface can bear. The sharp knife works because the small area gives a high pressure, and the dam base needs strength because depth gives a high pressure.
Lesson 5 Factors Affecting Pressure in Liquids: Depth, Density and Direction	We observed that at the same depth of 10 cm, salt water (1,030 Pa) gave a larger pressure than water (1,000 Pa), and cooking oil (920 Pa) gave a smaller pressure. We saw that containers of different shapes and amounts of water gave the same pressure at the same depth, and that streams from holes at the same depth were equal in all directions.	We learned that liquid pressure increases with depth and with density, but does not depend on container shape or the amount of liquid. At the same depth the pressure acts equally in all directions.	This helps explain the knife question by showing that pressure depends on particular quantities, not on how large or heavy something looks. For the knives, the force (40 N) is the same and the area is different, so the pressure is different. In liquids, depth and density decide the pressure. We still need to learn how to calculate liquid pressure and bring all the ideas together to fully explain why the sharp knife cuts.
Lesson 6 Applying $P = \rho gh$: Deriving the Liquid Pressure Equation and Calculating Pressure in Lake Victoria and Dams	We worked out the pressure of water at different depths: Lake Victoria 10 m = 100 kPa, 20 m = 200 kPa, 40 m = 400 kPa, and the base of a dam 30 m deep = 300 kPa. When the column base was 1 m ² or 2 m ² , the pressure was the same at the same depth.	Liquid pressure comes from the weight of the liquid above. Using $P = F/A$, $W = mg$, $m = \rho V$ and $V = Ah$, we get $P = \rho gh$. Pressure depends on the density ρ and depth h , not on the width of the lake or the amount of water. Pressure in kPa = pressure in Pa divided by 1,000.	The pressure in a liquid rises with depth because a deeper column has more weight pressing on each square metre. This is why a dam wall is thicker at the bottom (300 kPa at 30 m compared with 100 kPa at 10 m). It is the same idea as the knives: pressure is force on an area; with the knives the force (40 N) was the same but the area was different, so the pressures were different.
Lesson 7 Atmospheric Pressure and Its Effects	The water stayed in the inverted glass with the cardboard; the heated can collapsed when cooled in water; the straw held liquid when the top was closed; ears pop when travelling to higher places such as Nairobi; atmospheric pressure values fall from Mombasa 101 kPa to Kisumu 89 kPa, Nairobi 83 kPa and Mount Kenya 54 kPa.	Air has weight, so it exerts atmospheric pressure, about 101 kPa (101,000 Pa) at sea level, and it acts on every surface. Atmospheric pressure decreases with altitude. The air force on the glass mouth is $F = P \times A = 101,000 \times 0.004 = 404$ N, much more than the 2.5 N water weight. Air does not make one knife cut better than the other because it pushes equally on both.	The cardboard stays because the upward force of air (404 N) is much greater than the weight of the water (2.5 N). The can collapses because the pressure inside falls and outside air pushes it in. Ears pop because outside pressure falls as altitude rises. The straw works because atmospheric pressure pushes liquid up when pressure in the straw is lowered. For the knives, the same force (40 N) acts on different areas, so the difference is area, not air.

Lesson 8 Transmission of Pressure in Fluids: Pascal's Principle in Syringes, Hydraulic Jacks and Brakes	We observed that when we pushed the small syringe plunger, the water in the closed tube moved the large plunger. The small plunger moved farther than the large one. The jack data showed a 50 N effort on a 5 cm² piston and a 1,000 N load on a 100 cm² piston, both with a pressure of 100 kPa.	We learned that pressure applied to an enclosed fluid is transmitted equally in all directions (Pascal's principle). Since $F_1/A_1 = F_2/A_2$, the large piston, with 20 times the area, gives 20 times the force (1,000 N from 50 N). This is used in syringes, hydraulic jacks and brakes.	We can explain that the knives and the jack both depend on $P = F/A$. For the knives the force is the same (40 N) and the area is different, so the sharp knife gives 10 times more pressure (8 MPa vs 0.8 MPa). For the jack the pressure is the same (100 kPa) and the area is different, so the force is different (50 N vs 1,000 N). Area is the key in both cases.
Lesson 9 Applications and Consolidation: Siphon, Dams, Hydraulics and the Final Explanation of the Knife Phenomenon	We observed that Mama Wanjiru's sharp knife (edge 0.1 mm) cuts a tomato and nyama choma while the blunt knife (edge 1 mm) only squashes them, though the force is the same 40 N. We also observed a siphon moving water from a higher to a lower level, a dam wall that is thicker at the bottom, and a hydraulic model in which a small effort of 50 N gave a load of 1,000 N.	We learned that pressure is force per unit area, $P = F/A$, measured in pascals (Pa). For the same force, a smaller area gives a greater pressure. Liquid pressure increases with depth (300 kPa at 30 m in a dam compared with 100 kPa at 10 m); atmospheric pressure (about 101 kPa at sea level) helps a siphon work; and pressure in an enclosed fluid is transmitted equally, so $F_1/A_1 = F_2/A_2$ lets a jack lift a matatu.	Final Explanation: Mama Wanjiru pushes with the same force of 40 N on both knives. The sharp edge has an area of $5 \times 10^{-6} \text{ m}^2$ and the blunt edge $5 \times 10^{-5} \text{ m}^2$, so the sharp knife gives a pressure of 8 MPa and the blunt knife 0.8 MPa, which is 10 times less. Tomato and meat give way at about 2 MPa, so 8 MPa cuts and 0.8 MPa only squashes. Same force, different area, therefore different pressure; the difference between the knives is the area, not the force.

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